oxygen directly to form DOPA. The synthesis of melanin occurs in the melanocyte, and the reactions starting with tyrosine are shown in **Figure.7.7**.

DOPA is converted to DOPA quinone. A number of intermediates are formed giving melanin. The more common product is eumelanin, brown in appearance. However, in the presence of cysteine, pheomelanin can be formed (red to yellow). Melanin is formed primarily in the melanocyte, located in the inner layers of the skin where melanin and carotene blend to produce the skin colour as well as the colour in the eyes and hair (Figure.7.8).

7.4.2 Symptoms

In Oculo-cutaneous albinism, there is decreased pigmentation of skin and eyes, whereas in Ocular albinism, there is decreased pigmentation of only eyes and not the skin. The common symptoms include the following

- Absence of colour in the skin or iris of the eye
- Patchy and missing skin colour
- Crossed eyes (Strabismus)
- Light sensitivity (photophobia)
- Rapid eye movements (Nystagmus)



Figure 7.8. A woman affected with albinism

7.5 ALKAPTONURIA

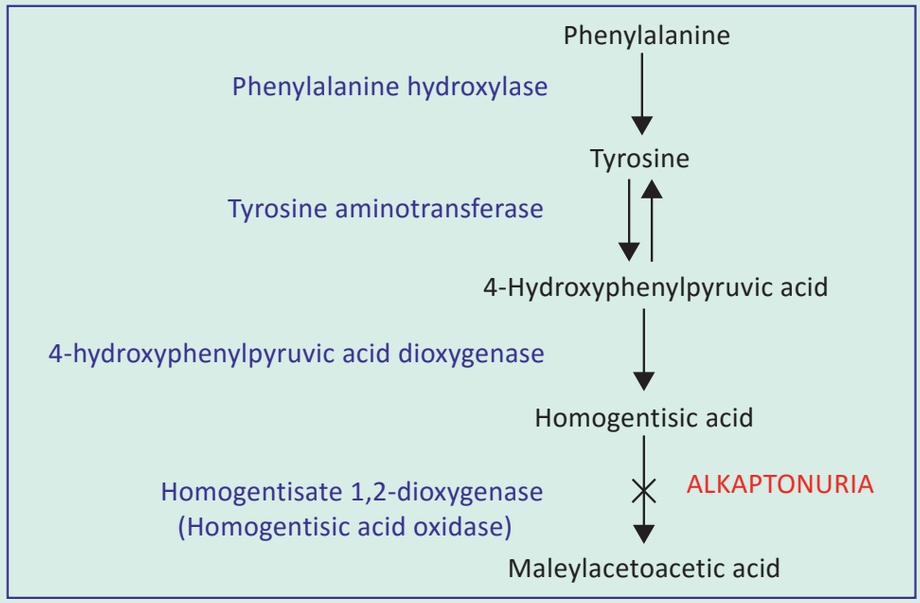


Figure 7.9 Metabolism of Phenyl alanine and Tyrosine

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